

## **A Validity Ladder for LLM-Simulated Populations**

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### **Author Note**

Data, code and every receipt file are at <https://github.com/pskeough/A-Validity-Ladder-for-LLM-Simulated-Clinical-Populations>.

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### Abstract

Behavioral researchers now prompt large language models with demographic personas and use the answers in place of survey respondents, and then check these simulated samples usually receive is a reading of individual answers. A simulated sample supports only the uses whose matching validity checks it has passed. The validity ladder is a protocol of a gate and four rungs. The gate asks how many repeated draws make a persona's score precise enough to read. The rungs ask whether a single draw is an answer pattern a person could give, whether gaps between groups match the population's, whether the level matches, and whether the items hang together as they do in people. Each rung has a statistic, a pass rule, and a use that a pass supports. We demonstrate the ladder on 28,800 PHQ-8 assessments from four models, read against the National Health and Nutrition Examination Survey. Persona means of a few draws are precise enough to read, and every model fails all four rungs above the gate. Single draws are less varied than real respondents at the same total score. The gap between low and high income is steepened in every model. Every model's population mean sits above the survey's after post-stratification. At level 4, two models have no general factor and the other two fail invariance across income. Survey respondents drawn through the same design pass the gate, levels 1 and 3, and the factor checks of level 4, and each planted failure trips the rung it targets. Level 2 rarely fails survey respondents and detects a doubled gap; certifying that a gap is kept needs more draws per persona and a population gap the survey measures precisely. The code, the corpus and a receipt for every reported number are public.

*Keywords:* silicon samples, large language models, simulated participants, validity, measurement invariance, PHQ-8, NHANES

## A Validity Ladder for LLM-Simulated Populations

Behavioral researchers now prompt large language models (LLMs) with demographic personas and treat the answers as *silicon samples*. Argyle et al. (2023) used such samples to reproduce vote choice and opinion by group, Park, Schoenegger, and Zhu (2024) used them to replicate classic experiments, and Lin (2026a) describe their use in piloting instruments before fielding them. For a study, a simulated sample costs little, reaches groups that are hard to recruit, and can be rerun whenever the design changes. Each of these uses assumes that the simulated sample stands in for a real one in the respect the study measures.

In practice the stand-in assumption is checked unevenly. Across eight social science journals, Desai, Card, and Jacobs (2026) found validation practice in LLM studies “inconsistent and limited.” Park et al. (2024) found GPT-3.5 answering with almost no variation where people vary, and Bisbee, Clinton, Dorff, Kenkel, and Larson (2024) found synthetic opinion distributions narrower than the survey distributions they imitate. Lin (2026b) proposed a workflow in which the validity a study needs rises with the claim it makes. Missing from the workflow is a fixed set of tests, run in a stated order with a pass rule and a population reference for each, that a researcher can apply to one model and report.

One persona from the corpus analysed here shows what reading individual answers misses. Prompted as a single Black woman on Medicaid with a household income under \$35,000, GPT-4o-mini returns a PHQ-8 total of 8 on its first draw, a plausible answer in the mild band. Over thirty draws the same prompt averages 10.3. Adults in the National Health and Nutrition Examination Survey (NHANES) who share her race, sex, income and marital status average 4.7. None of her single answers would show that the simulated population she belongs to sits well above the real one.

This paper gives a protocol for the missing check, the validity ladder (Figure 1). A gate first measures how many repeated draws make a persona’s score precise enough to

read. Above the gate, level 1 tests whether a single draw is an answer pattern a person could give, and level 2 tests whether gaps between groups match the population’s. Level 3 tests whether the population level matches, and level 4 tests whether the items relate to one another as they do in people. Each rung carries a statistic, a pass rule and the use a pass supports, so a simulated sample supports only the claims whose rungs it has passed. With the rules frozen and released as code, a researcher can run the ladder on their own sample and report the verdicts.

We apply the ladder to 28,800 PHQ-8 assessments from four models, read against NHANES. The assessments were first collected for an earlier audit (Keough, 2026), and every analysis reported here is rebuilt from the raw answers. Persona means pass the gate at 30 draws, and every model fails each rung above it. Real respondents drawn through the same design are rarely failed by any rung, and failures planted at known sizes are caught by the rung each targets. On three silicon-sample datasets released by Bisbee et al. (2024), Argyle et al. (2023) and Meister, Guestrin, and Hashimoto (2025), the ladder identifies which group gaps a model keeps, distorts or loses.

### **The Validity Ladder**

The ladder applies to any simulated sample in which a model answers a fixed instrument while conditioned on a persona. One draw is one call to the model and the answer vector it returns. A cell is one persona under one model and one prompt framing, and it holds  $k$  draws whose mean score is the persona mean. Figure 1 summarises the gate and the four levels, with the question each asks, the statistic it reads and the use a pass supports.

The gate tests the base premise of any simulated sample, whether the model gives a persona a score consistent enough to read. Each level above the gate supports a broader claim, moving from a single answer at level 1 to gaps between groups at level 2, the population’s level at level 3 and the structure of the instrument at level 4. A sample earns each claim separately, and the researcher reads off the claims their sample supports.

Without these checks, answers that look coherent are assumed coherent and a sample that looks typical is assumed free of bias.

A researcher deploys one model, so the ladder gives each model its own verdicts. Pooling several models into one verdict can hide errors that run in opposite directions, as the sex gaps in Section show. A pass means the test found no failure at the precision the data allow, and a level whose population reference is too imprecise to support any verdict records a stop in place of a pass or a fail. The gate fixes  $k$  for levels 2 and 3, and a failure at level 1 rules out single-draw uses only, so the other levels still run. The rules and thresholds were fixed before any of the datasets in Section were analysed. The thresholds are defaults that a user may change, and Supplement S7 gives the full specification, including estimators, bootstrap designs and multiplicity control.

### **Gate: Regeneration Stability**

The gate asks how many draws must be averaged before a persona's score is precise enough to read. Repeated draws of one persona differ because the model samples its answer, so each draw is the persona's own score plus draw-to-draw noise. A generalizability study (Brennan, 2001; Cronbach, Gleser, Nanda, & Rajaratnam, 1972) measures the two parts as variances,  $\sigma_p^2$  for the spread of scores between personas and  $\sigma_r^2$  for the spread of draws within one persona. Averaging  $k$  draws shrinks the noise in a persona mean, and its standard error and dependability are

$$\text{SE}(k) = \sqrt{\sigma_r^2/k}, \quad \phi(k) = \frac{\sigma_p^2}{\sigma_p^2 + \sigma_r^2/k}. \quad (1)$$

The standard error is the typical distance between a persona mean and the score the model would give over unlimited draws. Dependability, from 0 to 1, is the share of the variation among persona means that reflects real differences between personas.

Persona means of  $k$  draws may be read when  $\text{SE}(k)$  is at most 0.25 reference standard deviations in each framing, and the recommended  $k$  brings it to 0.125 standard deviations. On the PHQ-8 these tolerances are 0.98 and 0.49 points. A single draw may be

Figure 1

*The Validity Ladder*

| Level and question                                                                                                                      | Unit read                                                                                | Statistic and pass rule                                                                                                                                                                                                                                                                                                                                                                                                                                              | Comparator                                                                              | A pass allows                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LEVEL 4</b><br><b>Structural fidelity</b><br>Does the scale hang together the way it does in people, and the same way across groups? | <b>All draws</b> of one model and framing, as one population; personas resampled         | One-factor model on polychoric correlations; congruence with the NHANES loadings; invariance across sex, race and income.<br><b>Pass:</b> in both framings, a general factor (every loading $\geq .30$ , first-to-second eigenvalue ratio $\geq 3$ ), congruence $\geq .95$ at its lower 90% limit and loading RMSD $\leq .10$ at its upper limit, and every invariance step agreeing with NHANES at margin .08. The factor within personas is reported, not scored. | NHANES 2005–2018 loadings and invariance verdicts                                       | Treat the total as one score that means the same in every group                                                                                 |
| <b>LEVEL 3</b><br><b>Population calibration</b><br>Is each group's level right?                                                         | <b>Persona means</b> of $k$ draws, reweighted to each NHANES group's makeup              | Simulated minus NHANES mean PHQ-8, for all adults and for nine groups by race, sex and income.<br><b>Pass:</b> every 90% interval within $\pm 2$ points (0.2 SD and 1 point also reported). The share scoring 10 or more is a second row.                                                                                                                                                                                                                            | NHANES 2005–2018 group means (2021–2023 beside them)                                    | Use the simulated level as a stand-in for the population's                                                                                      |
| <b>LEVEL 2</b><br><b>Subgroup fidelity</b><br>Are the gaps between groups the right size and direction?                                 | <b>Persona means</b> of $k$ draws, differenced within pairs that differ on one attribute | Ratio of the simulated gap to the NHANES gap, with a 98.6% interval (Bonferroni over seven contrasts).<br><b>Pass:</b> the interval lies within 0.75 to 1.25.<br>Each contrast is kept, not kept (interval outside 0.75 to 1.25) or unresolved (interval overlapping it). Not read: no population gap; reference too imprecise to certify kept.                                                                                                                      | NHANES 2005–2018 gap with the other persona attributes held equal (crude gap beside it) | Read the size and direction of group differences                                                                                                |
| <b>LEVEL 1</b><br><b>Individual coherence</b><br>Does a single draw answer like one person?                                             | <b>Single draw:</b> one answer vector, with inference clustered by persona               | Person fit of the item pattern under a graded response model fitted to NHANES, compared with adults at the same total.<br><b>Pass:</b> the shares of draws below the NHANES 5th and above its 95th percentile, each divided by 5%, have 90% intervals within 0.67 to 1.5.                                                                                                                                                                                            | NHANES 2005–2018 adults with the same PHQ-8 total                                       | Use a single draw as one case, such as an exemplar or a test item<br><i>A failure rules out single-draw uses only. Levels 2 to 4 still run.</i> |
| <b>GATE</b><br><b>Regeneration stability</b><br>Are persona scores stable enough to read?                                               | <b>Repeated draws</b> of one persona, within one model and framing                       | Variance of scores between personas and between draws; dependability of a mean of $k$ draws.<br><b>Pass:</b> $k$ is the fewest draws giving $SE \leq 0.125$ reference SD in each framing (minimum rule 0.25 SD). A single draw passes with $SD \leq 0.25$ reference SD. Dependability is reported as persona separation.                                                                                                                                             | None: draws are compared with each other                                                | Read persona means of $k$ draws at levels 2 and 3                                                                                               |


 The gate fixes  $k$ , the number of draws averaged into each persona mean, for levels 2 and 3. Apart from this, every level runs whatever the result below it.

*Note.* Read from the gate at the bottom to level 4 at the top. Each row gives the question a level asks, the unit it reads, its statistic and pass rule, what it is compared with, and the use a pass supports. The gate sets  $k$ , the number of draws averaged into each persona mean, for levels 2 and 3. A level-1 failure rules out single-draw uses only, so levels 2 to 4 run whatever level 1 finds. Transgender and multiracial personas have no NHANES group of their own and never enter a group comparison.

read as the persona’s score only when the draw-to-draw standard deviation is itself at most 0.25 reference standard deviations. Dependability is reported beside every verdict and required ( $\phi \geq .80$ ) only for uses that compare or rank individual personas, because a faithful simulator inherits the population’s small differences between cells. Drawn through the persona design, real NHANES respondents reach a dependability of .70 at 30 draws. A pass supports reading the persona mean of  $k$  draws as the model’s answer for that persona, and it says nothing about whether the answer is correct.

### Level 1: Individual Coherence

Level 1 asks whether a single draw looks like an answer sheet a real person could give, and whether the draws are as varied as real people. Real respondents with the same total score tend to endorse the common symptoms before the rare ones, and a minority answer in unusual patterns. Person fit measures how typical an answer pattern is for its total. A graded response model (Samejima, 1969) is fitted to the population reference with survey weights, and every respondent and every draw receives the standardised person-fit statistic  $l_z^*$  (Drasgow, Levine, & Williams, 1985; Sinharay, 2016; Snijders, 2001), which is low for an unusual pattern and high for an unusually regular one. Each draw is then compared with real respondents who have the same total score.

The *misfit ratio* is the share of a model’s draws more unusual than 95% of those respondents, divided by the 5% expected, and the *overfit ratio* is the share more regular than 95% of them, divided by 5%. Both ratios equal 1 in a real population, and a ratio of 0.2 means the model gives a fifth as many such patterns as people do. A model passes in a framing when the 90% intervals of both ratios lie between  $1/\tau$  and  $\tau$ . The tolerance  $\tau$  is the smallest of 1.5, 2 and 3 that covers how far every real subgroup by sex, race and ethnicity, income and survey cycle clearly departs from 1 when scored against the whole reference, with a floor of 1.5 set by the positive control (Section ). On NHANES,  $\tau = 1.5$ . A model with too few patterns in both tails is called *compressed*. A pass supports using single draws as individual cases, such as exemplar vignettes or test items for a screener,

and a compressed failure rules out uses that depend on the spread of real presentations.

## Level 2: Subgroup Fidelity

Level 2 asks whether the gap between two groups, such as low- and high-income adults, has the population's size and direction. The simulated gap  $g$  compares personas that differ on one attribute and match on the rest, such as two personas identical except for income. The population gap  $\gamma$  is computed the same way in the survey, among people who match on the other attributes, and the crude survey gap is reported beside it. The statistic is the ratio  $\rho = g/\gamma$ . A ratio of 1 reproduces the population gap, 2 doubles it, 0 erases it, and a negative ratio reverses it. Its interval is a Fieller interval, the standard interval for a ratio of two estimates (Fieller, 1954), widened for the number of contrasts tested on each model, which gives 98.6% intervals for seven contrasts.

A contrast is KEPT when its whole interval lies between 0.75 and 1.25, NOT KEPT when the interval lies wholly outside that range, and UNRESOLVED when it overlaps it. For a contrast that is not kept, the ratio gives the direction of the error, with ratios above 1.25 steepened, ratios from 0.25 to 0.75 attenuated, ratios near zero missing and negative ratios reversed. Two stops are checked before any verdict. A contrast is not read when the survey cannot tell its population gap from zero, and it is not read when the survey measures the gap too imprecisely for any simulated gap to be certified kept, unless the contrast is already clearly not kept. A model passes level 2 when every contrast read is kept, fails when any is not kept, and is otherwise unresolved. A pass supports reading the size and direction of group differences from the simulated sample, and it says nothing about either group's level.

## Level 3: Population Calibration

Level 3 asks whether the simulated sample has a group's level, such as the mean PHQ-8 score of women or of low-income adults. A persona grid is designed and not sampled, so before the comparison the personas are reweighted to the real mix of people in each group. Each persona cell  $c$  in a group  $G$  is weighted by its share  $p_c$  of the group in the



reference, and the residual is

$$R_G = \sum_{c \in G} p_c (s_c - y_c), \quad (2)$$

where  $s_c$  is the persona mean and  $y_c$  the mean of real respondents who match that persona (Holt & Smith, 1979). A residual of +2 means the simulated group scores 2 points higher than the real group. Its interval combines the draw error within cells with a design-based jackknife of the survey.

A row passes at tolerance  $\delta$  when its 90% interval lies inside  $\pm\delta$ , an equivalence test by two one-sided tests (Lakens, Scheel, & Isager, 2018; Schuirmann, 1987). The researcher names  $\delta$  as the smallest difference that matters for the use. By default it is half a reference standard deviation, about 2 points on the PHQ-8, and 0.2 standard deviations is the strict alternative. A model passes when every group row passes. The share scoring at or above the instrument's cut point is a second row, because a correct mean can hide a wrong prevalence. A pass supports using the simulated level as a stand-in for the group's level, within  $\delta$ , for the prompt, persona grid and survey period tested.

#### **Level 4: Structural Fidelity**

Level 4 asks whether the items of the instrument hang together as they do in people. In real respondents the eight PHQ-8 items rise and fall together, so the total works as one depression score, and it works the same way in different groups. All draws of one model in one framing are treated as one population, with personas as the resampling unit. A one-factor model is fitted to the item correlations (Olsson, 1979) and checked against three rules. R1 requires a general factor, one shared dimension behind the items, with every loading at least .30 and the first eigenvalue at least three times the second, or the reference's own value where the reference falls short. R2 requires the items to load on that factor in the same pattern and with the same strength as in people, read as Tucker's congruence of at least .95 (Lorenzo-Seva & ten Berge, 2006) and a root mean square difference in loadings of at most .10, each at the conservative end of its 90% interval. R4 requires the factor to work the same way across sex, race and income wherever it does in

people, read on the RMSEA of each invariance step (Savalei, Brace, & Fouladi, 2024) against a margin of .08.

A model passes level 4 when it passes all three rules in every framing. Passing R1 and R2 supports scoring the total as one score with item weights like people’s. Passing R4 adds that a group comparison of the total means the same in the simulation as in people, so passes at levels 2 and 3 are read beside it. A diagnostic (R3) also reports whether the factor holds among draws of one persona, as it does among people who share a demographic profile.

### Worked Example

#### Data

The worked example uses the corpus of Keough (2026), in which language models answered the eight-item Patient Health Questionnaire depression scale (PHQ-8; Kroenke et al., 2009) as demographic personas. Four models, GPT-4o-mini, Gemini-3-Flash, DeepSeek-V3 and GLM-4.7, generated the corpus through OpenRouter between 28 December 2025 and 11 January 2026. The persona grid crosses race (Asian, Black, Hispanic, multiracial, White), gender identity (cisgender and transgender women and men), income (three levels) and relationship status (married, single), for 120 personas. Income was stated as a dollar-and-insurance label, from “Low (<\$35k, Medicaid)” to “High (>\$250k, Concierge)”. Each persona answered 30 times under each of two framings, one listing its attributes as fields (clinical) and one giving them as a first-person self-description (narrative). A shared system prompt cast the model as a “Clinical Simulation Engine” answering from “the statistical likelihood of symptoms for this specific demographic intersection in the US population”. No sampling parameter was set, so every draw ran at the provider’s default decoding. The corpus holds 28,800 assessments, of which 28,799 have a valid PHQ-8. Calls lost to API outages were resent in January under the same prompt, and every rung gives the same per-model verdicts without them (Supplement S3). Supplement S1 prints the prompts, and the generation code is released with the data.

**Table 1***Ladder Verdicts per Model in the Worked Example*

| Model          | Gate: draws needed<br>(minimum / rec.) | Level 1<br>single draws | Level 2<br>income gap ratio | Level 3<br>overall residual | Level 4<br>structure |
|----------------|----------------------------------------|-------------------------|-----------------------------|-----------------------------|----------------------|
| DeepSeek-V3    | Pass (3 / 9)                           | Fail, compressed        | Fail, 1.91                  | Fail, +4.6                  | Fail, no gene        |
| Gemini-3-Flash | Pass (3 / 10)                          | Fail, compressed        | Fail, 5.07                  | Fail, +2.6                  | Fail, R2 and         |
| GPT-4o-mini    | Pass (4 / 13)                          | Fail, compressed        | Fail, 2.16                  | Fail, +4.7                  | Fail, no gene        |
| GLM-4.7        | Pass (6 / 21)                          | Fail, too few atypical  | Fail, 5.03                  | Fail, +2.1                  | Fail, R2 and         |

*Note.* Gate: draws needed for a persona mean’s standard error to reach 0.25 (minimum) and 0.125 (recommended) NHANES standard deviations, the larger of the two framings. Level 2: ratio of the simulated to the standardised NHANES gap between low- and high-income personas, where 1 reproduces the gap. Level 3: post-stratified simulated minus NHANES mean PHQ-8, in points. Level 4: R2, loadings stronger than people’s; R4, the factor differs across income where it does not in NHANES. Full per-level tables are in Supplement S4.

The population reference is the 36,274 adults in NHANES 2005–2018 who answered all eight PHQ-8 items, analysed with the survey’s weights and design (Chen et al., 2018; Johnson et al., 2013; National Center for Health Statistics, 2018). Income is the ratio of family income to the poverty line, banded below 1.3, from 1.3 to 3.5, and above 3.5, and each persona’s income label is mapped to one band. Forty-eight of the 120 personas have a matching group in the survey. NHANES records sex and not gender identity, and it does not identify multiracial adults on their own, so transgender and multiracial personas enter levels 1 and 4, which read the whole simulated population, and no group comparison. NHANES 2021–2023 and other reference choices are sensitivity analyses in Supplement S3.

## Results

Table 1 gives every model’s verdict at each rung. All four models pass the gate and fail every level above it.

**Gate.** No model can be read from a single draw. The draw-to-draw standard deviation is 1.25 to 2.24 points in every model and framing, above the 0.98-point tolerance. Averages of 30 draws pass in every model, and the minimum rule needs 3 to 6 draws. Two draws of one persona fall in different PHQ-8 severity bands 35.2% of the time [33.6, 36.8]. In a separate decoding control on 12 personas, setting GPT-4o-mini’s temperature to 0 cut its draw variance to 12.5% of the default without separating personas any better (Supplement S1). The worked-case persona’s 30 clinical draws run from 7 to 16, and her first draw of 8 sits in the mild band while her mean of 10.3 sits in the moderate band.

**Level 1.** No model passes level 1, in either framing or at any tolerance up to 3. At matched totals NHANES puts 5% of adults in each tail of person fit, and the models put 0.0% to 1.0% in the unusual tail and 0.0% to 4.3% in the regular tail. Three models are compressed, and GLM-4.7 is short of unusual patterns alone. The models also lean on mood. At matched totals they score 0.16 to 0.51 points higher than people on depressed mood and 0.22 to 1.11 points lower on fatigue, one of the physical symptoms people report first. They reuse answer sheets as well, giving two different personas the identical answer vector 4.6 to 39 times as often as two NHANES adults with the same total. Each of the worked-case persona’s 60 draws is a plausible human pattern, and together they are less varied than real people at her severity.

**Level 2.** Every model fails level 2 on income (Figure 2). The gap between low- and high-income personas is 1.91 (DeepSeek-V3) to 5.07 (Gemini-3-Flash) times the population gap. Of the 12 per-model income contrasts, 10 are steepened, and the survey measures the other two gaps too imprecisely to certify kept. The sex gap divides the models. GPT-4o-mini and DeepSeek-V3 keep a quarter to two fifths of the population’s sex gap (ratios 0.23 and 0.41), GLM-4.7 doubles it (2.06), and Gemini-3-Flash is unresolved. Pooled over the four models, the ratio is 0.97, a near match produced by opposite errors. No racial contrast is kept in any model. Among adults matched on income, sex and marital status, the survey cannot tell the Hispanic–White gap from zero and measures the

**Figure 2**

*Level 2 in the Worked Example: Verdict for Each Model and Group Gap*

|                             | GPT-4o-mini                                 | Gemini-3-Flash                                 | DeepSeek-V3                                    | GLM-4.7                                        |
|-----------------------------|---------------------------------------------|------------------------------------------------|------------------------------------------------|------------------------------------------------|
| <b>Women – Men</b>          | <b>0.23×</b><br>missing or attenuated       | <b>1.17×</b><br>unresolved                     | <b>0.41×</b><br>missing or attenuated          | <b>2.06×</b><br>steepened                      |
| <b>Low – High income</b>    | <b>2.16×</b><br>steepened                   | <b>5.07×</b><br>steepened                      | <b>1.91×</b><br>steepened                      | <b>5.03×</b><br>steepened                      |
| <b>Low – Middle income</b>  | <b>2.08×</b><br>steepened                   | <b>8.38×</b><br>steepened                      | <i>not read</i><br><i>survey too imprecise</i> | <b>7.51×</b><br>steepened                      |
| <b>Middle – High income</b> | <b>2.25×</b><br>steepened                   | <i>not read</i><br><i>survey too imprecise</i> | <b>2.57×</b><br>steepened                      | <b>2.21×</b><br>steepened                      |
| <b>Black – White</b>        | <b>−0.76×</b><br>reversed or missing        | <i>not read</i><br><i>survey too imprecise</i> | <i>not read</i><br><i>survey too imprecise</i> | <i>not read</i><br><i>survey too imprecise</i> |
| <b>Hispanic – White</b>     | <i>not read</i><br><i>no population gap</i> | <i>not read</i><br><i>no population gap</i>    | <i>not read</i><br><i>no population gap</i>    | <i>not read</i><br><i>no population gap</i>    |
| <b>Asian – White</b>        | <b>0.42×</b><br>unresolved                  | <b>1.52×</b><br>unresolved                     | <b>0.74×</b><br>unresolved                     | <b>1.82×</b><br>unresolved                     |

kept (none here)
  not kept
  unresolved
  not read

*Note.* Each tile gives the simulated PHQ-8 gap as a multiple of the NHANES 2005–2018 gap among adults matched on the other persona attributes, so  $1\times$  reproduces the survey gap. A gap is kept when its 98.6% interval lies between  $0.75\times$  and  $1.25\times$ , not kept when the interval lies wholly outside that range, and unresolved when it overlaps it. For a gap that is not kept, the second line names the direction of the error. A tile is not read when the survey cannot tell its gap from zero or measures it too imprecisely for any simulated gap to be certified kept. No model keeps any gap. Supplement Figure S1 shows every interval.

Black–White gap too imprecisely to certify kept, and GPT-4o-mini’s Black–White gap is still not kept. The worked-case persona does not enter level 2 on her own, because a gap is read across pairs of personas.

**Level 3.** Every simulated population is more depressed than NHANES. After reweighting, the overall residual is +2.1 points for GLM-4.7, +2.6 for Gemini-3-Flash, +4.6 for DeepSeek-V3 and +4.7 for GPT-4o-mini. All 40 model-by-group residuals are positive, and no model passes all ten groups at 2 points. In every model the largest residual is in the low-income band. The failure holds under all eight reference specifications, including NHANES 2021–2023, and only whether the national row passes depends on the reference chosen (Supplement S3). Prevalence errs differently from the mean. In the low-income band 55.5% of simulated respondents score 10 or more, against 13.6% of real adults. In the middle and high bands the simulated share is too low although the mean is too high, and DeepSeek-V3’s overall prevalence looks correct only because its low-income excess cancels these shortfalls. The worked-case persona’s mean over both framings is 10.0, 5.3 points above the 4.7 of real adults in her cell.

**Level 4.** No model passes level 4. DeepSeek-V3 and GPT-4o-mini have no general factor, with first-to-second eigenvalue ratios of 1.09 to 1.73 against 7.47 in NHANES, so their PHQ-8 total is not one score. Gemini-3-Flash and GLM-4.7 have a factor with the human pattern of loadings (congruence .990 to .998), and the loadings are stronger than people’s. Gemini-3-Flash fails R2 in both framings and GLM-4.7 in the narrative framing, and both fail invariance across income, which holds in NHANES. In these two models 48% to 64% of item variance lies between personas, against 3.5% between demographic cells in NHANES, and draws of one persona share no general factor. Their factor is built from differences between persona means, and the uniformity within personas is the same one level 1 found.

**Summary.** Read one at a time, the answers of every model look like believable patients, and averages of a few draws are stable enough to read. None of the four models

supports a claim beyond that stable score. They compress the variety of individual answers, exaggerate the income gradient, raise every group’s level of depression and lose the structure of the scale. No single answer shows any of these errors, and each rung of the ladder exposes one of them.

### **Controls: Real People and Planted Failures**

Every model fails every rung above the gate, so the ladder itself has to be shown to pass real people and to catch failures of a known size. The controls build pseudo-models out of real survey respondents. Each replicate splits the NHANES adults at random into two halves within the 48 persona cells, uses one half as the reference, and fills the corpus design from the other half, with 120 personas, two framings and 30 draws, where each draw is a real respondent from the persona’s cell. A panel of four such pseudo-models stands in for the four language models. Failures are then planted into the same respondents, each aimed at one rung, by making the items independent, shifting every persona mean, doubling the income gap, compressing answer patterns toward the typical one, and making three items independent within the low-income group. Run on the real corpus with the full reference, the control code reproduces all 303 published rung numbers.

**Real people pass, and planted failures are caught.** Real respondents pass the gate and level 3 in every reading, level 1 in 92.9% of replicates, and R1 and R2 of level 4 in every replicate, and level 2 fails them in no reading (Table 2). Each planted failure trips the rung it targets. Independent items fail level 1, R1 and R2 in every replicate, shifted samples fail level 3 in every reading, compressed samples fail level 1 in every replicate, non-invariant samples fail R4 in all 25 replicates, and a doubled income gap fails level 2 in 92.5% of per-model readings. Two rungs also react to failures they do not target. A sample shifted by 2.1 points passes the gate in only 28.6% of readings, because more depressed respondents vary more between draws, and R2 fails compressed and non-invariant samples, because both change the size of the loadings. Over a further 200 replicates with graded doses, the distortions the language models show at levels 1, 3 and 4 lie beyond the dose

**Table 2**

*Verdicts for Seven Simulator Types Built From Survey Respondents*

| Simulator              | Gate  | L1   | L2 fail | L3    | R1    | R2    | R4 fail    | R4 unres. | L4  |
|------------------------|-------|------|---------|-------|-------|-------|------------|-----------|-----|
| Real respondents       | 100.0 | 92.9 | 0.0     | 100.0 | 100.0 | 100.0 | 4.0 (25)   | 96.0      | 0.0 |
| Independent items      | 100.0 | 0.0  | 0.0     | 100.0 | 0.0   | 0.0   | 0.0 (5)    | 0.0       | 0.0 |
| Shifted, +2.1          | 28.6  | 72.9 | 0.0     | 0.0   | 100.0 | 100.0 | 0.0 (5)    | 100.0     | 0.0 |
| Shifted, +4.7          | 0.0   | 15.7 | 0.7     | 0.0   | 100.0 | 100.0 | 0.0 (5)    | 100.0     | 0.0 |
| Income gap doubled     | 100.0 | 84.3 | 92.5    | 85.4  | 100.0 | 100.0 | 0.0 (5)    | 100.0     | 0.0 |
| Compressed             | 100.0 | 0.0  | 0.0     | 100.0 | 98.6  | 1.4   | 0.0 (5)    | 100.0     | 0.0 |
| Non-invariant (income) | 100.0 | 2.9  | 0.0     | 100.0 | 100.0 | 0.0   | 100.0 (25) | 0.0       | 0.0 |

*Note.* Percentage of readings that pass, except the columns marked fail or unresolved. Gate, L2 and L3 are read per pseudo-model (280 readings per type); L1, R1 and R2 on one pseudo-model per replicate (70 replicates). R4 is read at margin .08 on the number of replicates in parentheses; it gives no verdict without a general factor. L2 passes in no reading of any type; its remaining readings are unresolved. L4 is the level-4 pass (R1, R2 and R4 in both framings). Shifted types raise every persona by the stated points; the doubled gap raises low-income personas only; non-invariant makes three items independent within the low-income group. Construction, seeds and per-step results: `paper_brm/analysis_brm/intermediate_panel` in the release.

that each rung detects in 95% of replicates (Supplement S5). Under the persona design, real respondents' misfit ratio averages 1.15, and their misfit tail passes at  $\tau = 1.5$  in 99.4% of replicates, which sets the level-1 floor.

**What level 2 and R4 need to certify a pass.** Two rungs need more data to certify a pass than to detect a failure. A second simulation repeated the panel over 60 replicates on grids of 48, 144 and 432 persona cells. Level 2 fails a faithful simulator in at most 3.3% of per-model readings and fails a doubled income gap in 97.5% to 100%. At 30 draws per persona, a faithful simulator's low-to-high income gap is never read KEPT, because the simulation's own error keeps its interval wider than the kept region. With no



simulation error, which is the limit of adding draws, the same gap is read KEPT in 95.0% and 98.3% of replicates on the 48-cell and the 144-cell grids. The Black–White and Hispanic–White gaps are never read KEPT on any grid, because NHANES measures them too loosely. A researcher who needs a kept gap therefore plans for more draws per persona and for a population gap the survey measures precisely, and a finer persona grid does not supply the second. R4 has the same property. At the .08 margin, 24 to 40 personas per group leave some invariance steps unresolved for real respondents in 24 of 25 replicates, while a planted invariance failure fails R4 in every replicate. A design that needs an R4 pass needs more personas per group than the corpus has.

### Three External Datasets

The ladder was applied without changing any rule to three silicon-sample datasets released by other authors (Figure 3). For each dataset we report which rungs the release allows and what the verdicts tell a researcher who would use that sample.

#### **Bisbee et al.: ChatGPT Feeling Thermometers**

Bisbee et al. (2024) gave ChatGPT (gpt-3.5-turbo, June 2023) the profiles of 7,530 respondents to the 2016 and 2020 American National Election Studies and asked, about 30 times under each of three prompts, for a feeling thermometer toward each of 11 groups. Every persona has a human twin, so the reference is the same respondents’ own thermometers, and the release supports the gate, level 2 and level 3. Our cleaning of the released file reproduces the authors’ analysis data and their published regression to six decimals. The gate passes for every thermometer, with at most 11 draws needed under the recommended rule, and level 2 fails under every prompt. Under the full prompt, 3 of the 77 contrasts are kept, among them the partisan gaps in feeling toward the Democratic and Republican parties (ratios 1.12 and 1.05), which are large gaps the survey measures precisely. The Republican–Democrat gap is steepened toward five other groups, and several gaps by sex and age are reversed, missing or attenuated. Level 3 passes at half a standard deviation for 7 of the 11 thermometers under the full prompt. For a researcher, these

**Figure 3**
*Level-2 Verdicts in the Worked Example and Three External Datasets*


*Note.* Bars give the share of group contrasts in each verdict, with counts printed inside segments wide enough to hold them; the column on the right counts the kept contrasts. Contrasts are per model in the worked example and OpinionQA, and per thermometer or outcome in the releases of Bisbee et al. and Argyle et al., which compare each persona with the respondent whose profile it was given. Verdicts use the same rule as Figure 2, with a Bonferroni correction within each model, prompt or run.

verdicts support using the sample for the partisan gaps toward the two parties and for the average level of most thermometers, and they rule out using it for partisan feeling toward other groups or for differences by sex and age. Bisbee et al. (2024) already read these data with caution, and the ladder places that caution on the specific uses that fail.

### Argyle et al.: GPT-3 Answering for Survey Respondents

In their Study 3, Argyle et al. (2023) gave GPT-3 each of 4,270 American National Election Studies respondents' answers to 11 of 12 interview questions and asked it to answer the twelfth. The study tested pattern correspondence, the criterion of their algorithmic fidelity under which simulated answers reproduce the relationships among

attitudes, demographics and behaviour that people show. The release holds one answer per respondent and outcome, so the gate and level 1 cannot run, and level 2 compares groups of the same respondents. Of 61 contrasts in the main run, 2 are kept, the Republican–Democrat gaps in ideology and in Trump’s two-party vote share (both 0.92). Nine are not kept, among them the Black–White gap in Trump’s vote share, which GPT-3 reproduces at about half its size (0.53). Most of the remaining contrasts are not read, because the survey’s subgroups are too small to measure their gaps. The verdicts support the partisan gaps, where the original claim is strongest, and leave the racial and other demographic gaps unsupported, so the claim of algorithmic fidelity holds within that narrower range of uses. Level 3 for this release is an illustration in Supplement S6, because it needs a use tolerance that the original authors did not state.

### **OpinionQA: What a Crude Reference Allows**

Meister et al. (2025) steered five models toward demographic groups on 100 contested questions from the Pew American Trends Panel and released the models’ answer distributions beside the Pew distributions for the same groups (Santurkar et al., 2023). The human side is each group’s distribution of answers to a question, built from unweighted counts, with no individual respondents. Without repeated draws of a persona or individual answer vectors, the gate, level 1 and level 4 cannot run. The model is steered on one attribute, such as being a Democrat, while the Pew groups also differ in age, income and every other attribute, so the reference is a crude gap and not the conditional gap the steering asks about. Each contrast is also averaged over questions with differing gaps, which leaves its interval wide.

Of the 154 contrasts across models, elicitation methods and steering methods, 1 is kept, 33 are not kept, 115 are unresolved and 5 are not read. The point ratio falls inside the kept region for 37 of them, so a comparison of point estimates would credit far more fidelity than the data establish. The ladder reports those contrasts as unresolved, and a researcher using these releases can claim little about group gaps in either direction.

## What the Three Datasets Show

Applied to other authors' samples, the ladder separates the uses a release supports from the uses it does not, and it reports where the human data are too thin to decide. Repeated draws of each persona under a fixed prompt make the gate and level 1 possible, and individual answer vectors make level 4 possible. Personas built from real respondents' profiles, as in the releases of Bisbee et al. (2024) and Argyle et al. (2023), give a reference that matches the simulated contrast. A large, individually measured reference makes a kept verdict reachable for the gaps that matter, while a crude, group-level reference leaves level 2 able to detect broken gaps and unable to certify faithful ones.

## Discussion

In the worked example and the three external datasets, every sample failed at least one check on a use that researchers make of such samples. Every finding drawn from a simulated sample therefore needs its own check. A group gap, a population level and a scale score are separate extrapolations, and a pass on one carries no information about the others. Researchers running persona studies should hold each extrapolation in critical doubt, audit it against the level that tests it, and treat a claim without that check as unsupported. The ladder makes the audit routine, and the controls show that it passes real respondents and detects failures of a known size.

Every level above the gate rests on human data. A national survey or a private validated dataset on the same instrument has to be in hand before the simulated sample is generated, because the reference fixes which personas can be compared, how many draws they need and which gaps the survey can measure at all. The cleanest reference comes from personas built on real respondents' profiles, as in the releases of Bisbee et al. (2024) and Argyle et al. (2023), where every simulated answer has a human twin.

For published silicon samples, the ladder narrows claims more often than it overturns them. Partisan gaps in the releases of Bisbee et al. (2024) and Argyle et al. (2023) hold, while most gaps by sex, age and race fail or cannot be read, so claims of

fidelity from these samples apply within partisan comparisons. The ladder leaves the substantive question to the human data. Its contribution is a stronger context for each claim and a screen against readings of a persona sample that the data do not support.

The worked example covers four base models under one prompt family, with one wording for each attribute, one persona per demographic cell, no age in the personas and the PHQ-8 items in a fixed order. Transgender and multiracial personas have no comparison group in NHANES and are not tested at levels 2 and 3. The corpus serves as a baseline against which grounded or fine-tuned simulators can be tested with the same ladder, and the code for every rung is released for that purpose.

## Conclusion

The validity ladder turns the question of whether a simulated sample can stand in for people into a set of checks, one for each use, each with a pass rule and a population reference. Applied to 28,800 PHQ-8 assessments from four models, it reads every model's persona means and fails every model on every use above them. Applied to three published silicon samples, it separates the uses each release supports from those it does not.

## Declarations

**Funding.** No funding was received for this work.

**Competing interests.** The author has no competing interests to declare.

**Ethics approval.** Not applicable. The study analyses public, de-identified NHANES microdata and text generated by language models. No human participants were recruited.

**Consent to participate and for publication.** Not applicable.

**Availability of data, materials and code.** The corpus, the persona registries, the original generation code for both framings, the analysis scripts and every receipt file are at <https://github.com/pskeough/>

**A-Validity-Ladder-for-LLM-Simulated-Clinical-Populations.** NHANES data are public from the National Center for Health Statistics, and the release includes the code

that extracts them.

**Author contributions.** Patrick Keough is the sole author and takes responsibility for all content.

**Use of generative AI.** Anthropic’s Claude models, accessed through Claude Code, were used to implement the analysis and verification scripts to the author’s specification, for LaTeX formatting and typesetting, and to draft prose from the author’s analysis reports and dictated text. The author designed the study, chose the estimands and the statistical specification, interpreted every result, and revised all text. All generated code was reviewed by the author, and every numeric claim in this paper is checked against its receipt by the verification scripts released with the data. The language models studied in this paper are the objects of the research and are described in the Worked Example section.

### Open Practices Statement

The data, materials and analysis code for this study are available at <https://github.com/pskeough/A-Validity-Ladder-for-LLM-Simulated-Clinical-Populations>. The analysis reported here is release commit `fd7ef46`. Supplement S2 maps every number in the body of the article to the file in that repository that prints it and the script that writes the file. The ladder’s rules were set on the worked example, run once on BFI-44 answers from three models (Jiang et al., 2024) against the Twin-2K-500 panel (Toubia et al., 2025), and frozen in `LADDER_SPEC.md` at commit `91a1b10` (2 October 2026), before any external dataset was analysed; the corpus analyses were not registered in advance. The decoding control’s design file, dated before its first call, is in the repository.

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